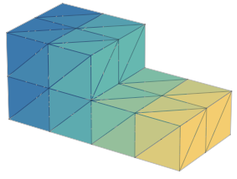
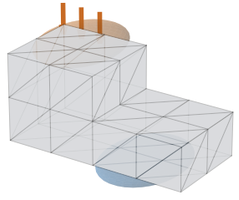
1 Spatial hypotheses





Material field **Load + support fields**

Material, support and load parameters

2 Integrate unknown loads

Possible load histories



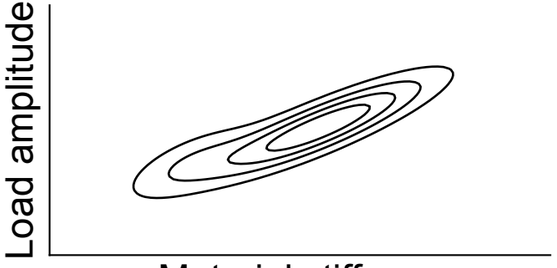
$$y = H_\theta a + d_\theta + \varepsilon$$

$$a \sim N(0, \Lambda)$$

$$C = H_\theta \Lambda H_\theta^T + R$$

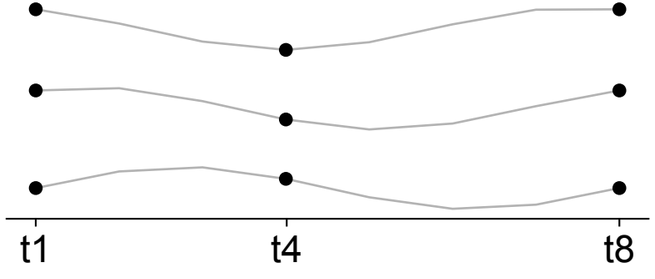
Exact Gaussian integration

3 Retain joint uncertainty



Keep coupled explanations

Observed surface tracks

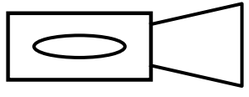


y

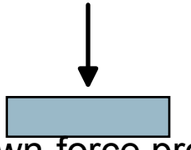
new evidence

4 Look or probe?

Minimize expected prediction error



Camera track

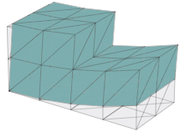


Known-force probe

Same target and measurement budget

5 Predict held responses

Hidden motion + changed loading



Prediction error and coverage